



Fig. 13: Typical SAR limit test model (courtesy: Compliance Engineering)

such public places as airports, railway stations and shopping malls. Their number is rapidly increasing in line with the growth in demand for mobile phones. Micro cell base stations emit less power (typically 5W) and their range is a few hundred metres (typically 300 metres). The EM radiation levels are expected to be much lower than the specified threshold.

Pico cells. Pico cells provide more localised coverage than micro cells. These are normally found inside buildings where coverage is poor or where there are a large number of users, such as airport terminals, train stations or shopping centres. They have lower power output than micro cells (typically 2 watts). The EM radiation levels are supposed to be even lower than in micro cells.

Biological effects from cellphone transmitting antennae. Exposure to very high levels of RF radiation can be harmful due to the ability of RF energy to rapidly heat biological tissues. Adverse effects (tissue damage) in humans could occur during exposure to high RF levels because of the body's inability to cope up with or dissipate the excessive heat that could be generated. The eyes and the testes are particularly vulnerable to RF heating because of the relative lack of blood flow in these organs to dissipate the excessive heat.

At relatively low levels of RF radiation exposure, the evidence of harmful biological effects is ambiguous

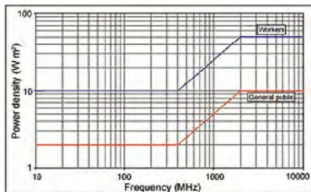


Fig. 14: Variation of ICNIRP specified reference level for power density as a function of frequency

ICNIRP based reference levels for exposure to time-varying electric and magnetic fields (unpublished raw values)				
For Occupational Personnel/Workers				
Frequency range	E-field strength (V/m)	H-field strength (A/m)	Equivalent plane wave power density P_{eq} (W/m ²)	Averaging period (min)
900 MHz	90	0.24	22.5	6
1800 MHz	127.28	0.339	45	6
2100 MHz	157	0.36	60	6
For General Public				
900 MHz	41.25	0.111	4.5	6
1800 MHz	58.24	0.167	9	6
2100 MHz	61	0.168	10	6

Fig. 15: ICNIRP-based field strength and power density levels at frequencies of interest

ous and unproven. Further research is needed to determine the effects and their possible relevance, if any, to human health.

Quantification of biological effect.

The rate at which RF energy is actually absorbed in the human body is called the specific absorption rate (SAR) – the fundamental dosimetric unit of RF power deposition. It is usually expressed in units of watts per kilogram (W/kg) or milliwatts per gram (mW/gm).

The average SAR (for whole body) is defined as total energy transferred to the whole body per unit time divided by total body mass. In MKS units:

$$\text{SAR} = \frac{\sigma E_{\text{in}}^2}{\rho} \text{ W/kg}$$

where 'σ' is the tissue conductivity (Siemens/m), 'E_{in}' is RMS value of the internal electric field intensity (V/m) and 'ρ' is the tissue density (kg/m³). Note that in the range of 80-100 MHz, SAR is maximum under these conditions (resonance). The resonance frequency shifts to 30-40MHz range when a man is in contact with conducting ground. Because of the resonance

phenomenon, RF safety standards are generally most restrictive at these frequencies.

In a biological body, SAR depends on several exposure parameters, viz. frequency, intensity and polarisation of EM radiation, as also on shape, size and electrical properties of the body. It is not readily measurable in living people. Usually, only research laboratories make SAR measurements because these are relatively difficult and require specialised equipment and conditions; e.g., implantable E-field probes and mathematical/physical models such as the ones shown in Fig. 13. SAR

measurement is more relevant for measuring EM radiation effect on parts of the human body from cellphones. For further information, you may refer to Armstrong Laboratory's dosimetry handbook and International Electrotechnical Commission's document IEC 62209-1 (2005) on the Internet.

EM radiation compliance levels. The National Council on Radiation Protection and Measurements (NCRP), USA, the Institute of Electrical and Electronics Engineers (IEEE) and the International Commission on Non-Ionising Radiation Protection (ICNIRP) have adopted a whole-body SAR value of 4 watts per kilogram (4W/kg) as a threshold level of exposure at which harmful biological effects may occur. Exposure threshold limits in terms of field strength and power density were then derived from this value. Under the circumstances, it is more practical and convenient to measure related radiated power density or E field or H field (averaged over a stipulated period for workers and general public) in the far-field region of interest at a cell site to ensure compliance with stipulated levels